Unit 13, Lesson 6: Interior Angles of a Regular Polygon

Benchmark

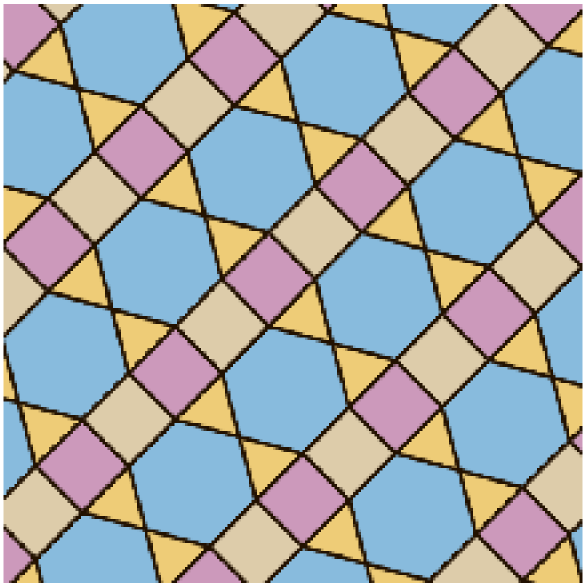
MA.8.GR.1.6 Develop and use formulas for the sums of the interior angles of regular polygons by decomposing them into triangles.

Learning Targets

- ☐ I can use formulas to find the sum of the measures of the interior angles of a regular n-gon.
- ☐ I can determine the measure of an interior angle in a regular n-gon.

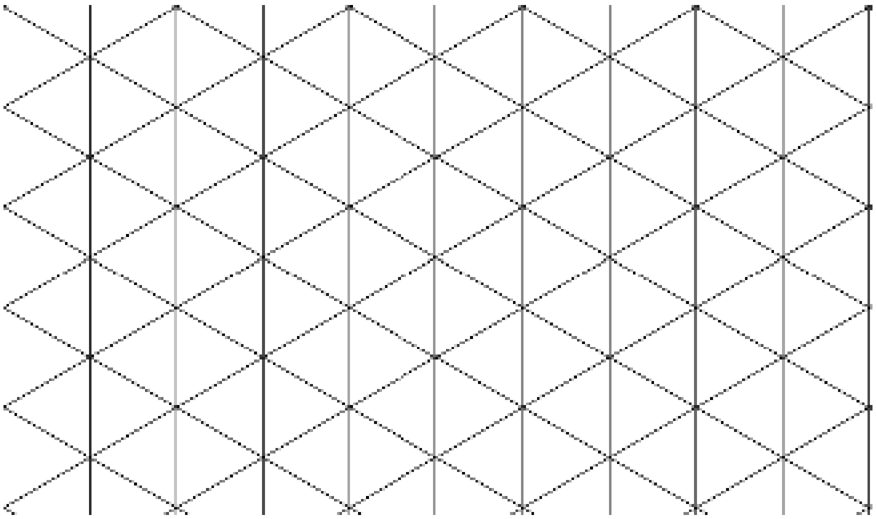
13.6.1 Warm-Up: Tiles and Regular Polygons

A tiling consists of covering a surface using copies of shapes, called tiles, without overlapping or leaving gaps. A picture of a tiling is shown.



1. Trace over and name each of the different regular polygons you notice in the picture.

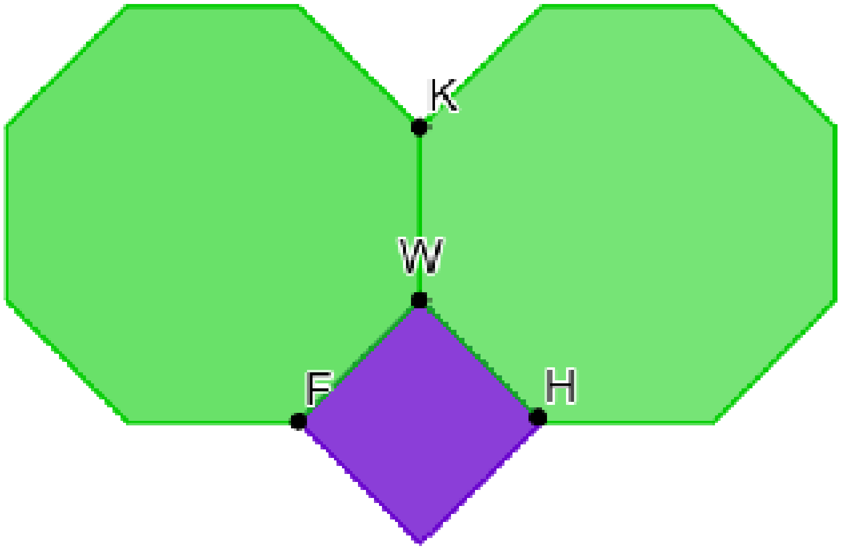
2. Use the grid to create a tiling.



13.6.2 Collaboration: Interior Angles of Regular Polygons

The **sum of the interior angles** of an n-gon can be found using the formula $\mathbf{180(n-2)}$, where n is the number of sides of the polygon.

1. **Jerrica plans to use a design similar to the one shown to tile her bathroom floor.**



a. **Work with your partner to find the interior angle sums of each polygon included in the tile design. Show your work in the space provided.**

Regular Octagon	Square

b. Work with your partner to determine the measure of each angle.

Measure of $\angle FWH$ = __________

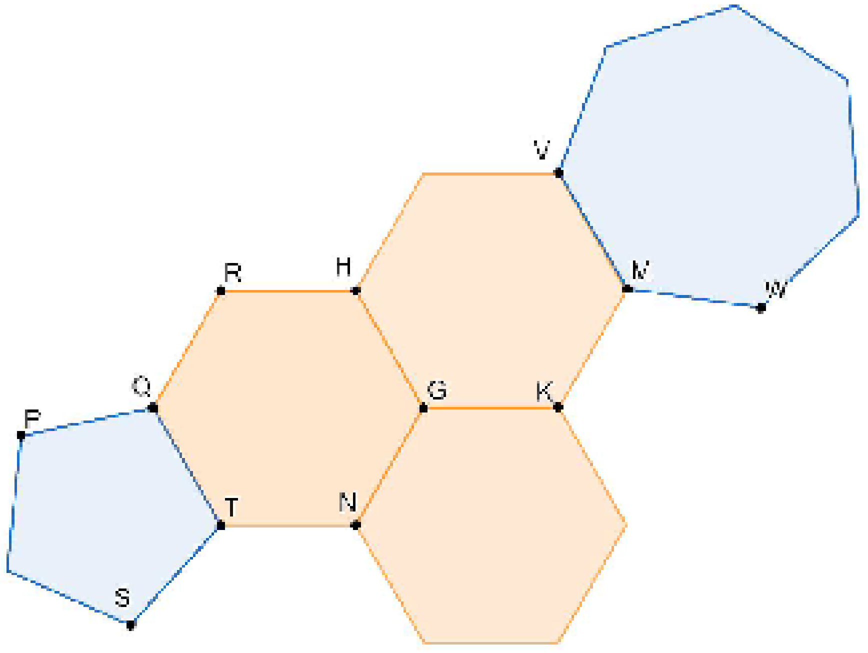
Measure of $\angle FWK$ = __________

Measure of $\angle HWK$ = __________

c. What type of angles are $\angle FWH$ and $\angle HWK$?

- o adjacent
- o complementary
- o supplementary
- o vertical

2. A diagram of regular polygons is shown.



Find the measure of each sum.

a. $m\angle STQ + m\angle NTQ =$ __________

b. $m\angle NGH + m\angle HGK =$ __________

c. $m\angle PQT + m\angle QRH + m\angle VMW =$ __________

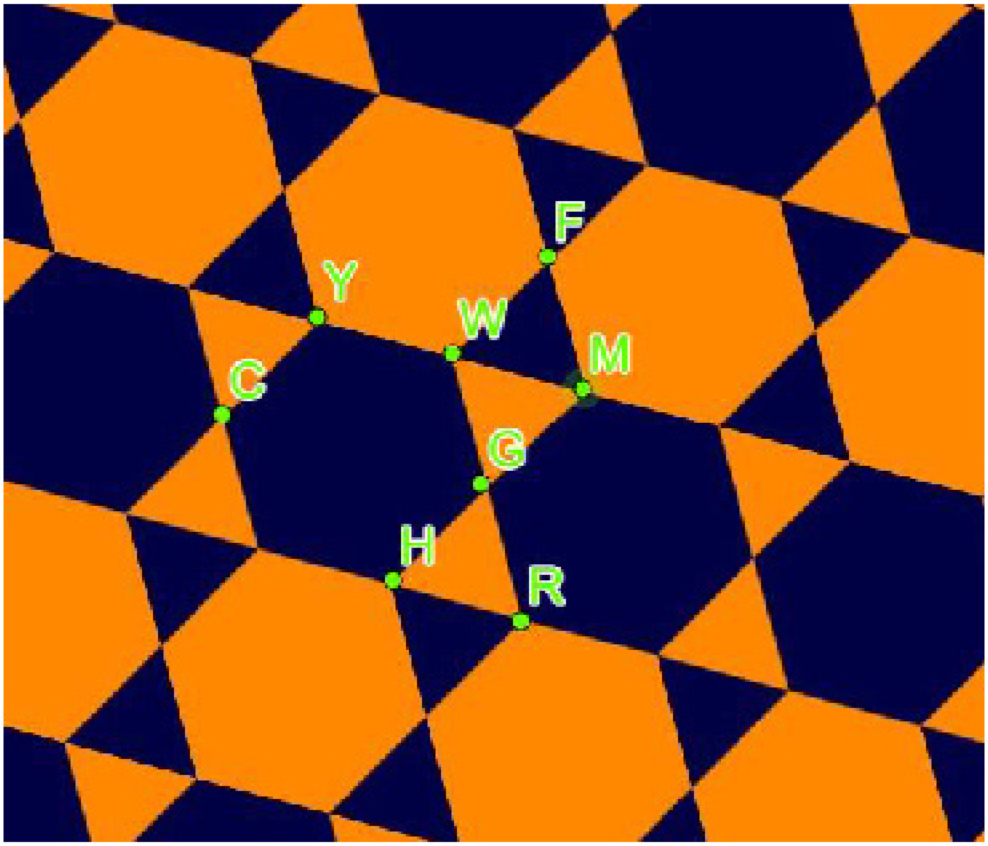
13.6.3 Guided Practice: Interior Angle Sums and Individual Angle Measures

1. **A regular polygon has 14 sides.**

   a. **What is the sum of the measures of the interior angles of this polygon?**

   b. **What is the measure of one interior angle of this polygon?**

2. **The tile pattern uses regular polygons.**



   a. **What regular polygons are in the tile pattern?**

b. **Find the measure of each.**

$m\angle FWY =$ ________

$m\angle HGW + m\angle HGR =$ ________

$m\angle MGR + m\angle GRH =$ ________

3. **The Pentagon is the headquarters building of the U.S. Department of Defense in Washington, D.C. Each of the 5 outside walls measures 921 feet, making it a regular pentagon.**

Curtis plans to build a scale model of the Pentagon.
What is the measure of the angle where any two outside walls join?

13.6.4 Your Turn!

1. What are the measures of each interior angle of a regular dodecagon, a polygon with 12 sides?

2. How many degrees greater is the measure of one interior angle of a regular nonagon (a polygon with 9 sides) than the measure of one interior angle of a regular octagon (a polygon with 8 sides)?

13.6.5 Wrap-Up: Error Analysis

1. Cathy is finding the measure of one interior angle of a regular pentagon. Her thinking and work are shown.

I can draw the diagonals from one vertex to form 4 triangles.

The sum of the measures of the interior angles is $4 \times 180° = 720°$.

The pentagon has five equal angles, so each angle measures $\frac{720°}{5} = 144°$.

Her solution is incorrect. Write a note to Cathy to explain the mistake she made. Then, provide her with the correct steps.

Means
I
Start
To
Acquire
Knowledge
Experience
Skills

